

Hutch-ALTTPR Tracker

User Guide

Version 1.1.2

A Link to the Past Randomizer — Auto-Tracking Tool

This is a simple guide for running the tracker. The tracker was built over many hours.

Disclaimer: This tracker is provided free of charge and is updated on a best-effort basis when issues are discovered. All source code is available at the GitHub repository linked below.

[GitHub Repository](#)

Quick Start Guide:	2
Prerequisite:	2
Install Hutch-ALTTPR Tracker	2
Windows Install:	2
UPGRADE	4
MacOS Install:	5
UPGRADE	5
Launching the Tracker	5
Tracker Launch Settings	7
CONNECTION:	7
DISPLAY:	7
SETTINGS:	10
Using the Tracker	13
Standard Randomizer	14
Map/Compass	15
Map/Compass/Small Key	17
Key Sanity	19
Other Mode	20
Using the Timer	22
Required Software Installation	24
QUSB2SNES Setup	24
RetroArch Setup (QUSB2SNES)	25
SNI Install	26
WINDOWS	26

MACOS	27
RetroArch Setup (SNI)	29
Troubleshooting	30

Quick Start Guide:

For best results here is the launch order that was tested.

- Launch QUSB2SNES or SNI
- Open the Emulator (RetroArch)
- Launch the game to the title screen
- Launch the various tracker components (map, Items, timer)
 - These should all show connected
- Start the game

Prerequisite:

- Install an Emulator that is compatible with websocket
 - Retroarch v1.2.2 or higher is preferred
- Install either SNI or QUSB2SNES
- Websocket for SNES (Either SNI or QUSB2SNES)
 - SNI (Works on Windows or MacOS)
 - <https://github.com/alttpr/sni/releases>
 - QUSB2SNES (Works on Windows or MacOS)
 - <https://skarsnik.github.io/QUsb2snes/>
- A Link to the Past v1.0 (JP) .sfc game
- Randomized game created
 - <https://alttpr.com/> Has instructions

Install Hutch-ALTTPR Tracker

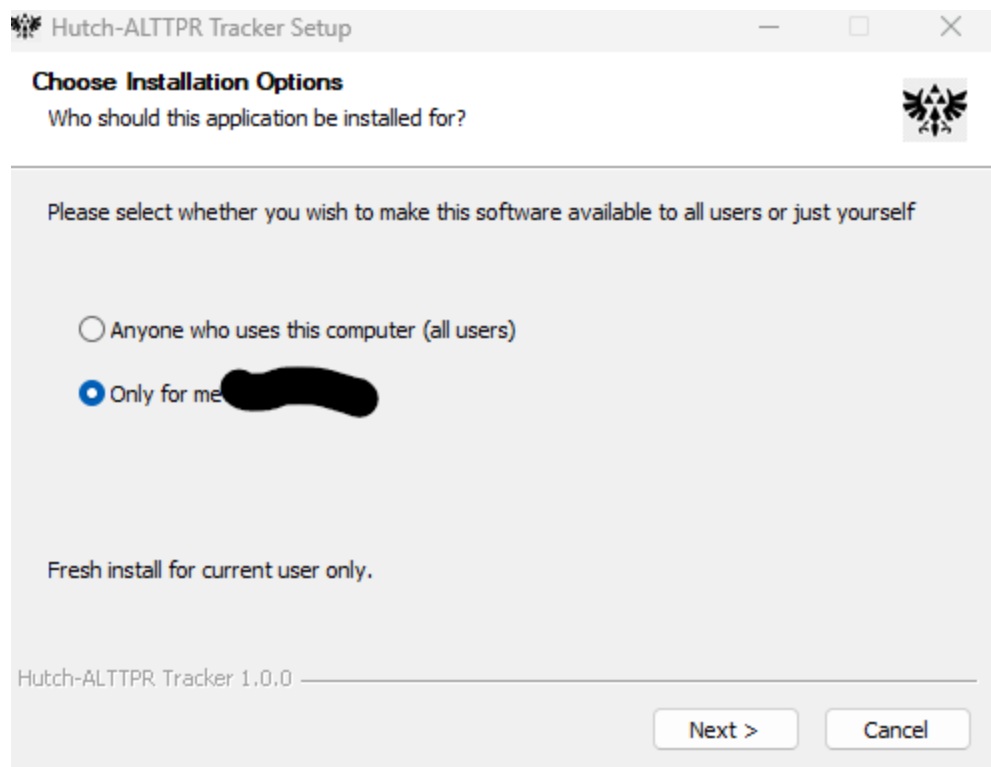
Windows Install:

Download the Hutch-ALTTPR Tracker from
<https://github.com/hutchch/ALTTPR-Tracker/releases>

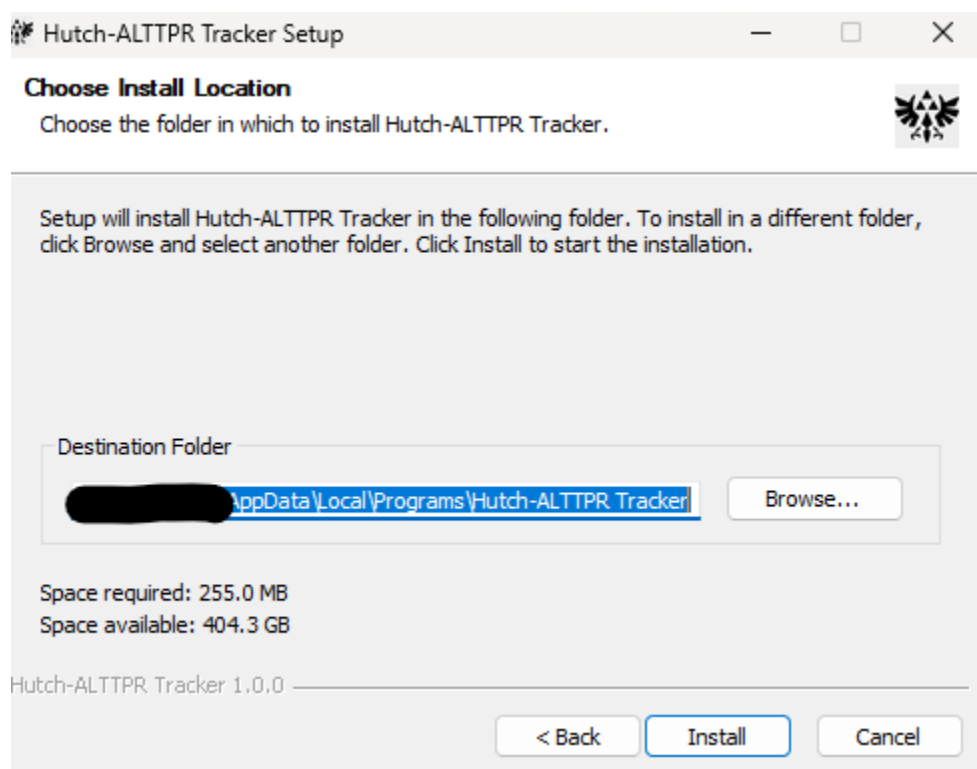
Extract the .zip and run the setup file.

Select either option

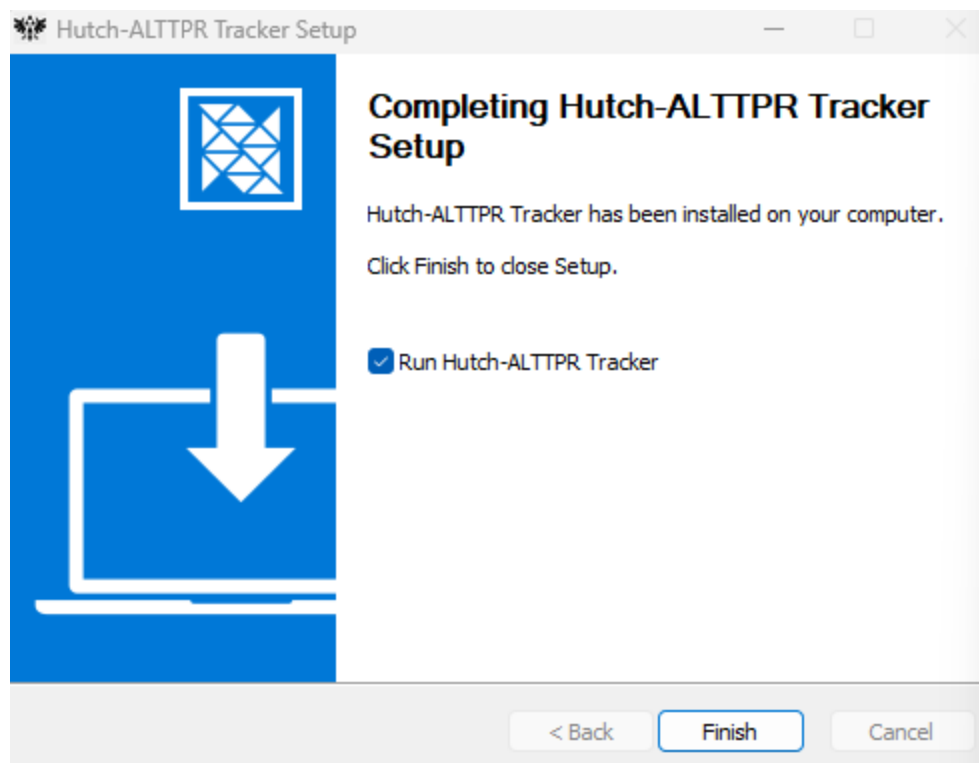
NOTE: Anyone option will require administrator privileges.



Select the destination folder:



Complete the setup

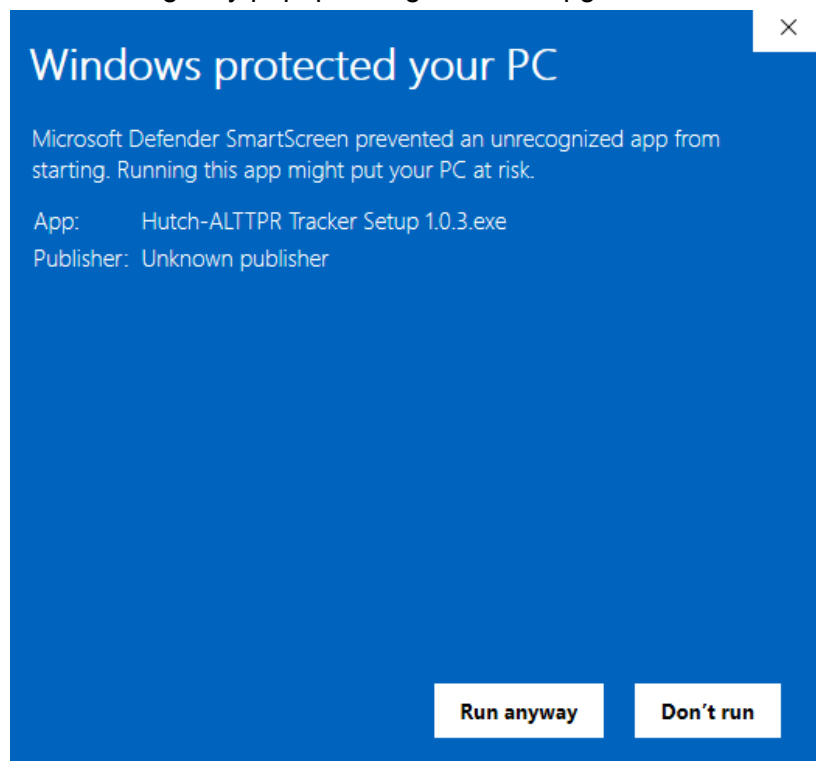


UPGRADE

To upgrade versions, proceed exactly the same as the install.

NOTE about Windows:

The following may popup during install or upgrade:



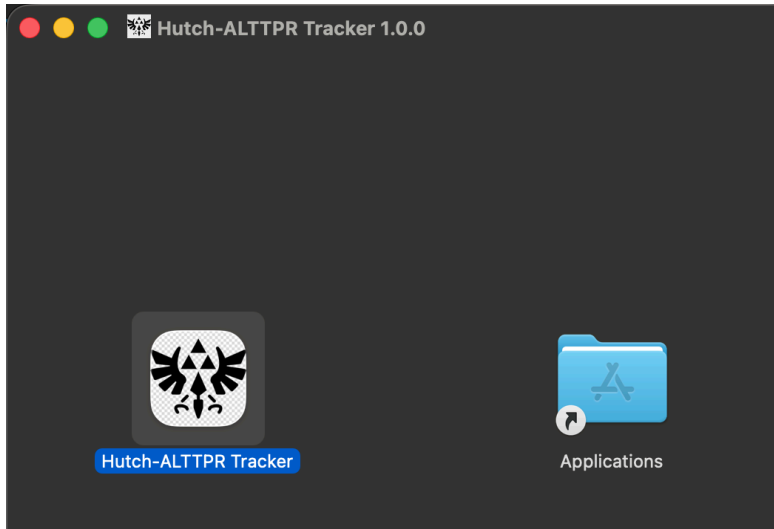
Select "Run Anyways" to upgrade.

The publisher certificate is an expensive purchase, and this is meant as a tool to help others. All the code is available in the main repository.

MacOS Install:

Download the Hutch-ALTTPR Tracker from <https://github.com/hutchch/ALTTPR-Tracker/releases>

Run the .dmg file and drag the program into the Application folder



UPGRADE

To upgrade versions, proceed exactly the same as the install.

Launching the Tracker

When Launching, select the game style settings and press the Launch buttons. (The Items/Map can be launched separately if the scaling needs to be adjusted).

✕ ALTTP Tracker

Version 1.1.1

CONNECTION

Host

Port

QUsb2Snes / SNI — default port 23074

DISPLAY

Item Scale

Tracker BG

Map Scale

SETTINGS

Dungeon Item Shuffle

Mode ☒ Standard ☐ Open

Enemizer ☒ Yes ☐ No

Ganon's Tower Crystal Requirement

Items Only

Map Only

Launch Both

TIMER

Number Color

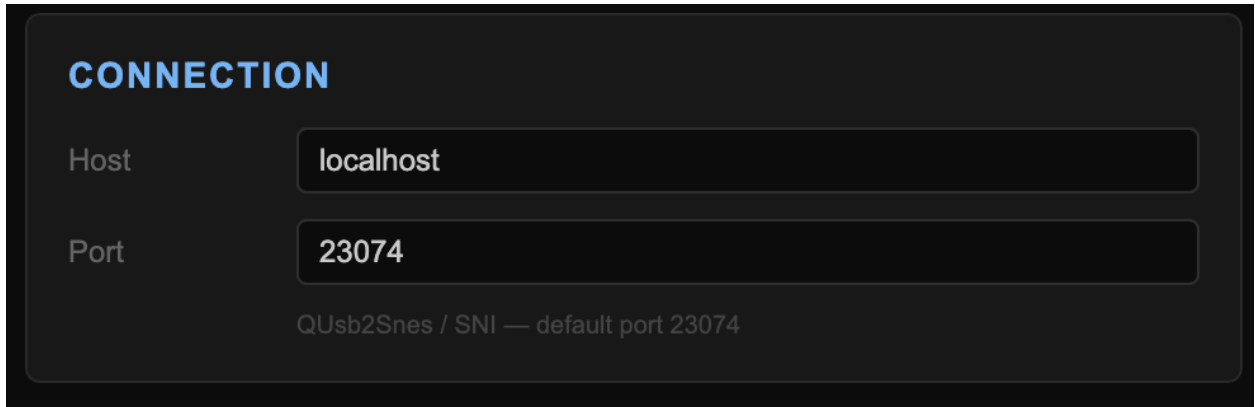
Background

🕒 Timer

Tracker Launch Settings

Select the settings for the game that was generated. Each setting has different logic, so for the best experience match the settings on the generated game.

CONNECTION:



CONNECTION

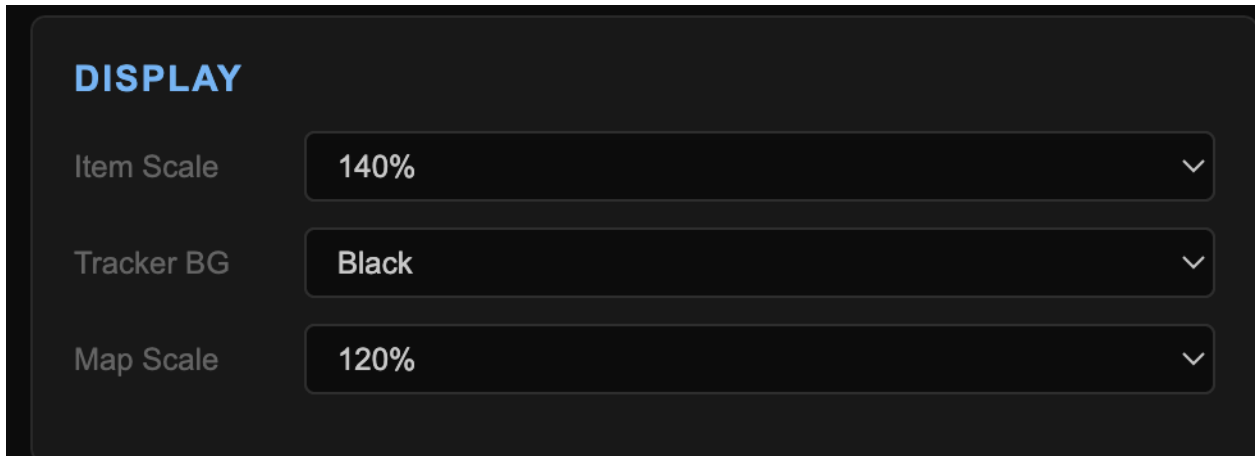
Host

Port

QUsb2Snes / SNI — default port 23074

The connection is SNI or QUSB2SNES socket information. The ports are the default.

DISPLAY:



DISPLAY

Item Scale

Tracker BG

Map Scale

Item Scale is the initial size of the Item Tracker. (The bottom bar does have resize buttons).

Tracker BG (Background) is the background color.

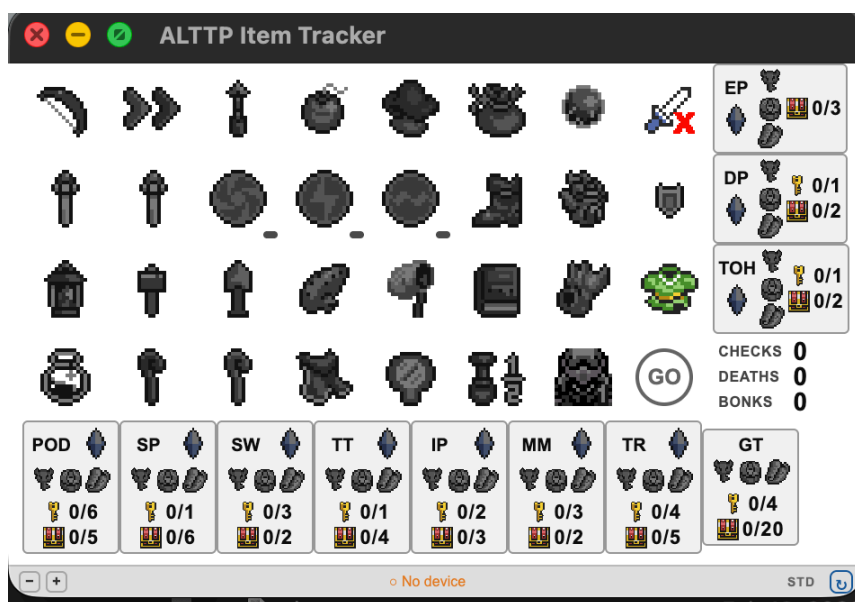
Black:



Grey:



White:



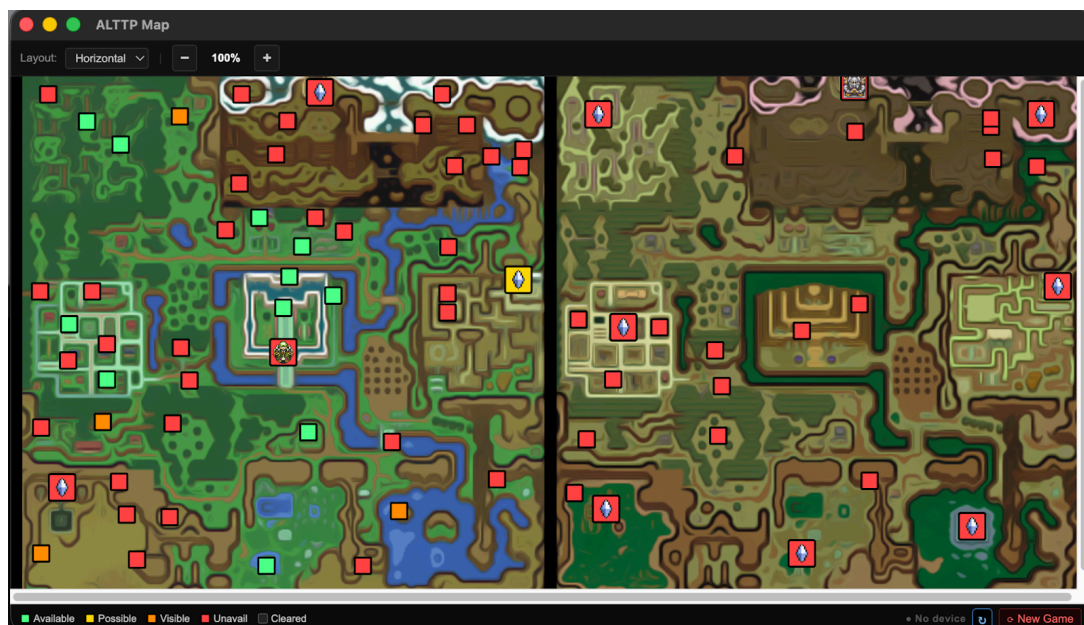
Transparent



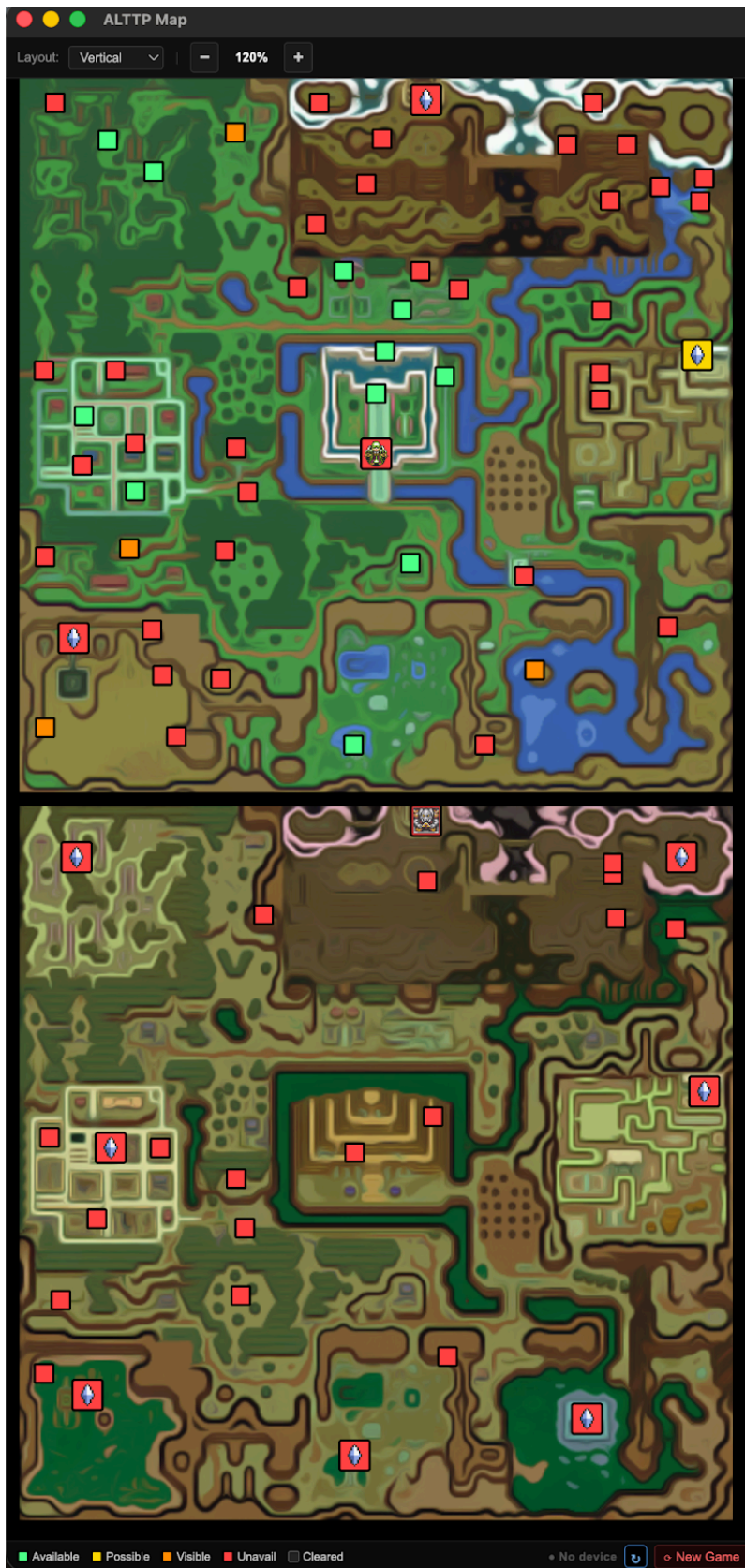
Map Scale is the initial size of the map

The Map can be rescaled after launch. Also Vertical and Horizontal views are available on the top "Layout"

Horizontal:



Vertical:

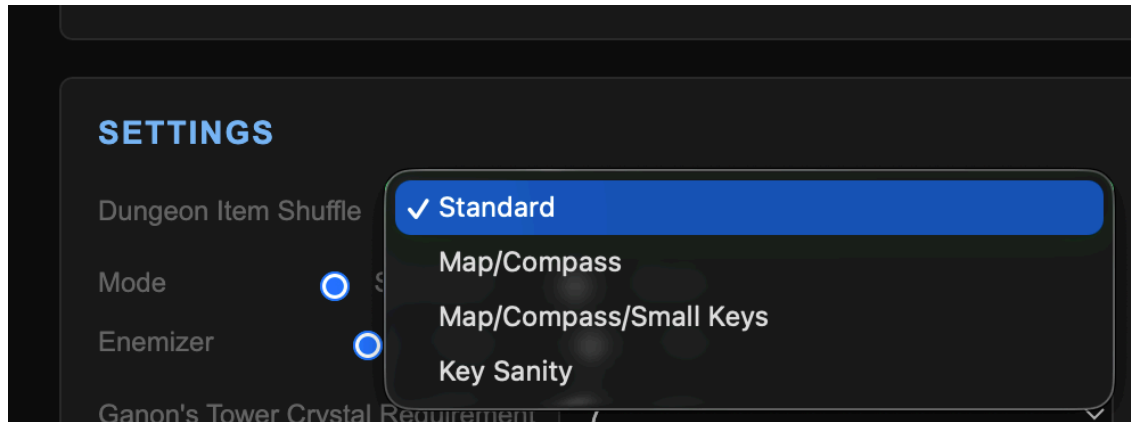


SETTINGS:

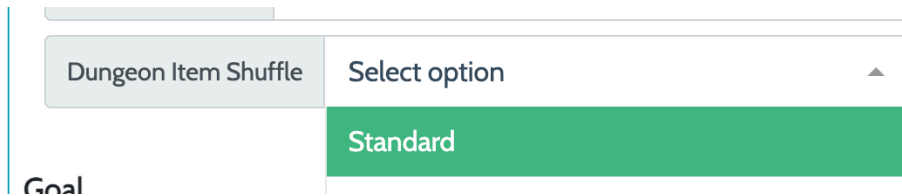
Settings are the game mode configuration for the seed.

Dungeon Item Modes:

Dungeon Item Mode is for the Dungeon Item Shuffle from the generated seed. Typical Randomizers should be standard. These settings correspond to the seed settings from alttpr.com



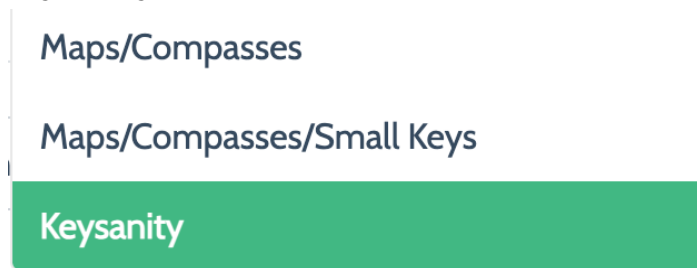
Standard: is for Standard settings in the Dungeon Item Shuffle from alttpr.com



Map/Compasses

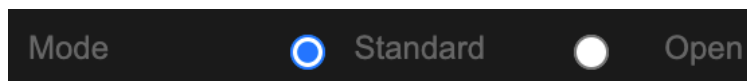
Map/Compasses/Small Key

Key Sanity



Mode:

Mode refers to the World state. Standard is the typical escape, and Open is without escape.



When Generating a Seed it is referenced as:

World State

Standard

The other options Open mode should be sufficient. See <https://alttpr.com/en/options> for information.

Enemizer:

This option is if enemies are random or shuffled.

Enemizer

☒

Yes

☐

No

If the seed was generated with None, then select No. If it was random or shuffled then select Yes.

Enemy Shuffle

Random

Ganon Tower Crystal Requirement:

This is the setting that the tracker will use to show if GT is green or not on the map. Match the "Open Tower" in the Goal settings.

Ganon's Tower Crystal Requirement

7

Goal

Goal

Defeat Ganon

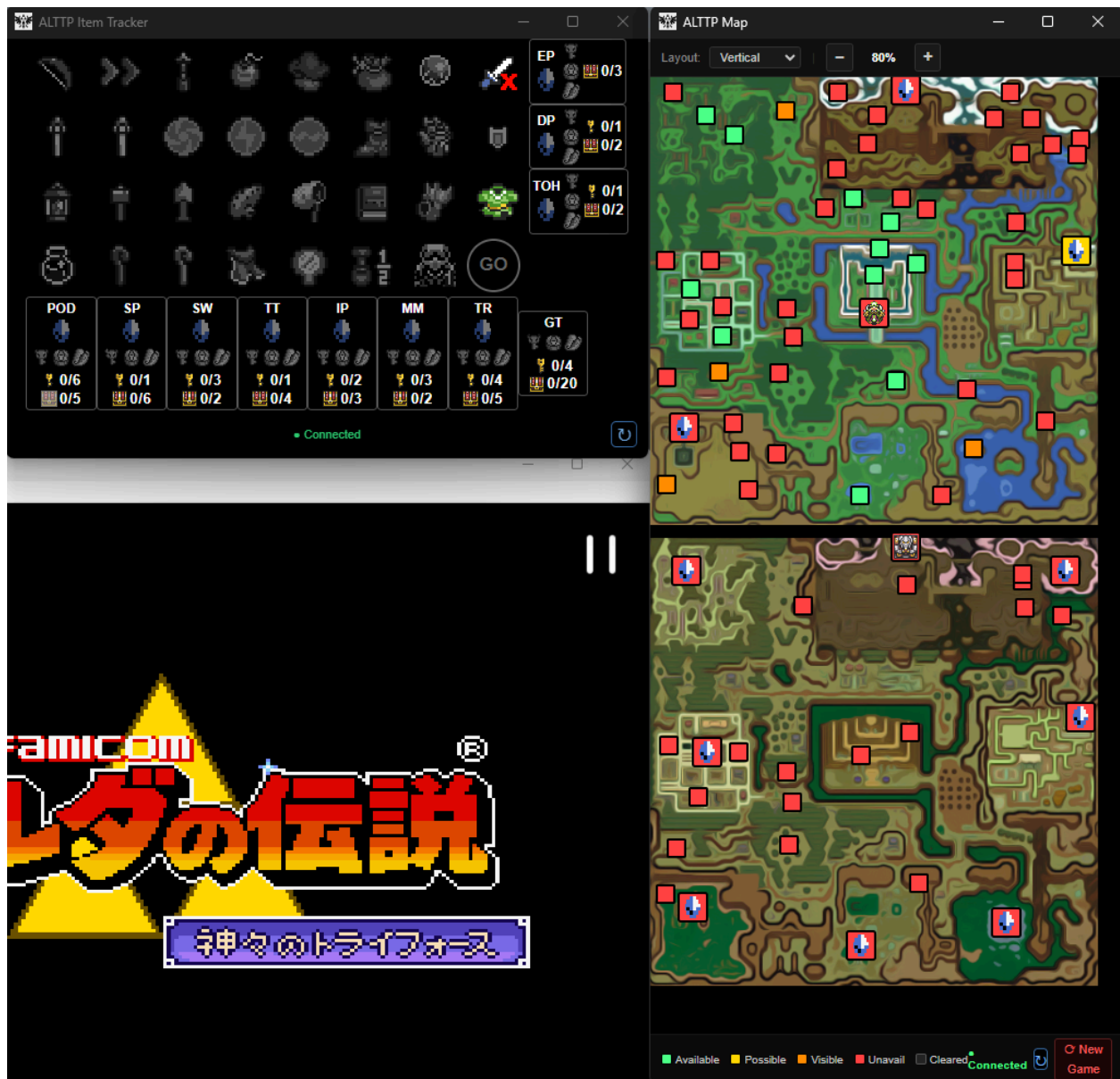
Open Tower

7 Crystals

Ganon Vulnerable

7 Crystals

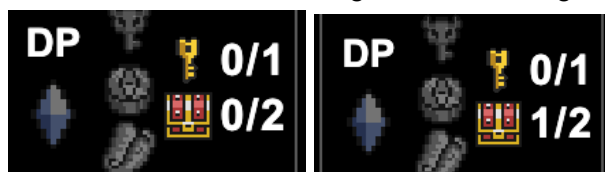
Once launched, the tracker should connect. If not, hit the reconnect button in the bottom right corner.



Using the Tracker

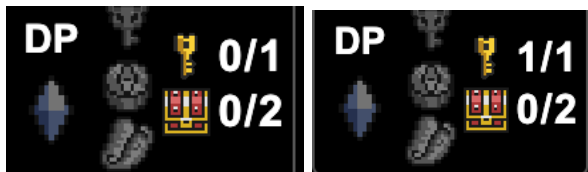
The tracker has various modes. The tracker was built with autotracking in mind. There may be a need for manual tracking. Prior to v1.1.2, everything but the dungeon chests and small keys can be manually set. With version 1.1.2 and newer the dungeon tracking can be manually, provided the websocket is not used (No Device status).

Left click the chest to change the values, right click will decrement the counts



NOTE: If the tracker is connected, Auto Tracking is preferred and this option is not available.

Small keys will work the same way, left click to increase the count, and right click to decrease.



NOTE: If the tracker is connected, Auto Tracking is preferred and this option is not available.

Standard Randomizer

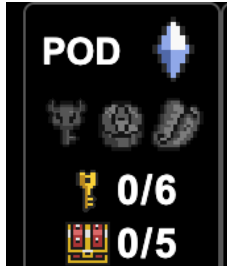
Standard mode is the most common seed. This mode will be the basic randomizer seed.

Dungeon Prize Marking

Click on the crystal to cycle through the various dungeon prizes. Since the most common is the Crystal, initially all dungeons are marked as the crystal so the user only needs to focus on the pendant and the red crystal markings



For manual use, the prize can be right clicked to show obtained. Right click the prize to show as obtained. A left click will override it.



Medallion Marking

Right Click the medallions to mark for Misery Mire or Turtle Rock



There is also a “BOTH” option if the same medallion opens both dungeons.

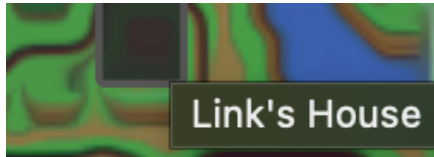


Map tracker:

The map will automatically clear the check when they are obtained.



Hovering the mouse over the check will provide the name



As with any tracker, the user can click on the checks to clear them or reset.

To reset all checks press the “New Game” and all checks will reset.



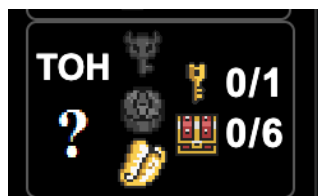
If the check has been cleared in the game, it will automatically clear. This is handy if the user cleared a check, but wants to reset the map to see what checks were skipped.

Map/Compass

In this mode the map and compass are shuffled. The item counts in the dungeon will increase by 2 over the standard mode. The item tracker is launched with “?” images for the dungeon prizes. The user must change the dungeon prize.



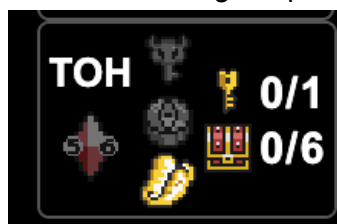
When a map is found, the Item tracker will light up the map icon for the dungeon it was discovered for.



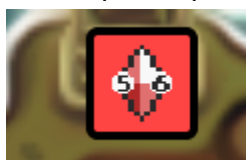
The map will display the map icon for that dungeon showing the map is found and the prize can be seen in the game play map



Click on the dungeon prize to change to the correct one from the game



The map will update with the appropriate icon



Map/Compass/Small Key

In this mode the small keys, maps and compasses are shuffled. Increasing the dungeon item count. As with the Map/Compass mode, the item tracker is launched with “?” images for the dungeon prizes. The user must change the dungeon prize.

EP	?	0/5
?	?	0/5

DP	?	0/1
?	?	0/5

TOH	?	0/1
?	?	0/5

CT	?	0/2
CHECKS	0	
DEATHS	0	
BONKS	0	

POD	?
?	0/6
?	0/13

SP	?
?	0/1
?	0/9

SW	?
?	0/3
?	0/7

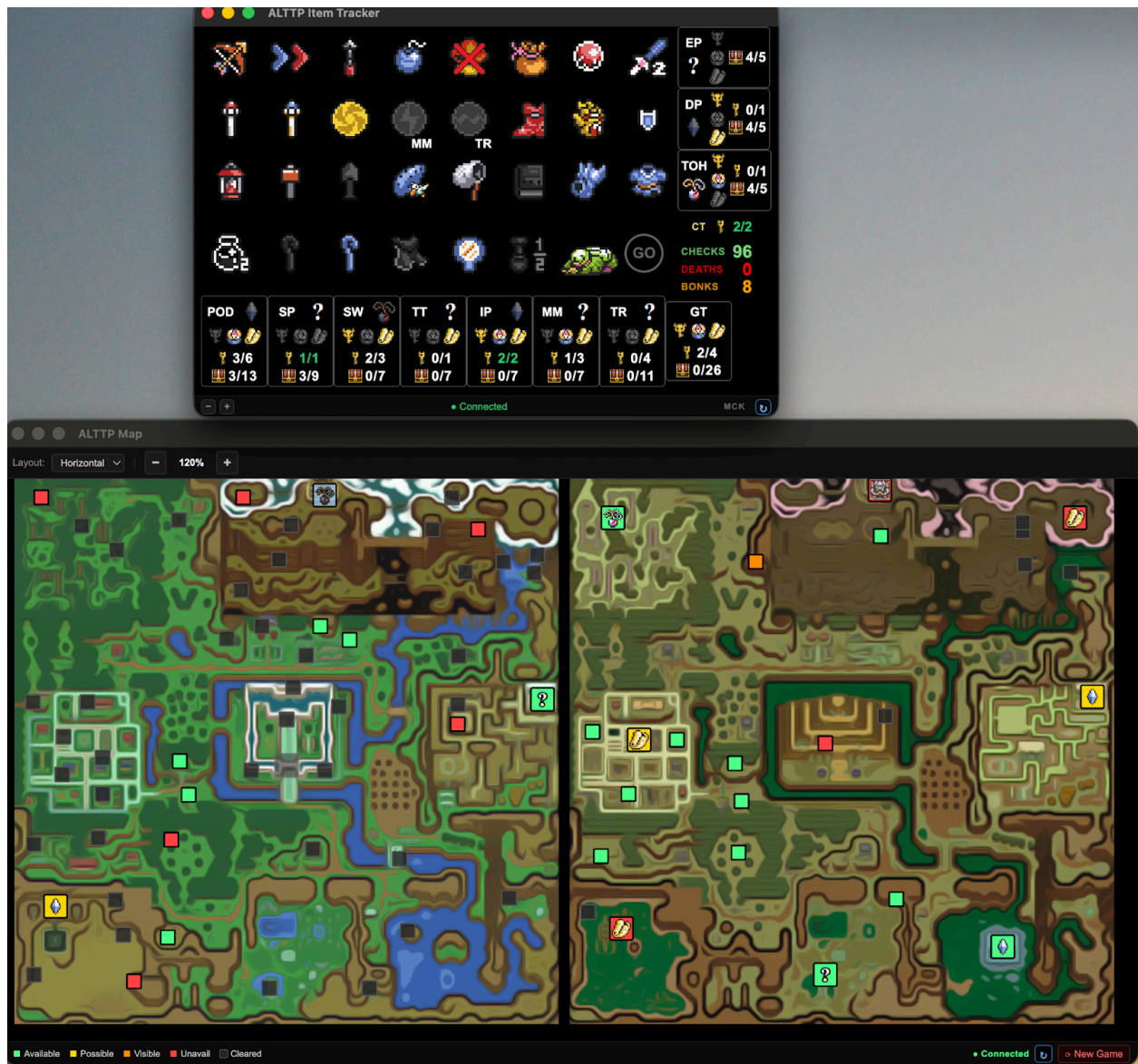
TT	?
?	0/1
?	0/7

IP	?
?	0/2
?	0/7

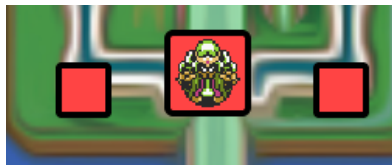
MM	?
?	0/3
?	0/7

TR	?
?	0/4
?	0/11

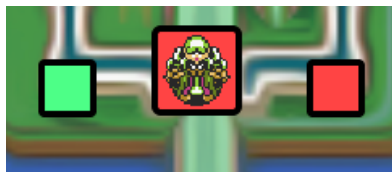
GT	
?	0/4
?	0/26



Map/Compass/Small Key, and Key Sanity also introduces 2 more checks on the map. The are Castle Tower "CT"



To access the first check, the tracker checks for the required items to get past the barrier.



The first check is available with either the Master Sword or the Magic Cape. The second Check requires 1 small key that is tracked.



This is a dark room, so lamp is also required for it to turn green. If the player can navigate the dark rooms, then the check can be cleared.

Aga1 will require the second CT key before the tracker turns green.



Because each dungeon can be completed with key logic, the map will show the dungeons green with the required items are discovered. Keys and Big Keys are tracked in the Item Tracker

When all small keys are discovered, the keys will turn green



Key Sanity

Key sanity refers to big keys, small keys, compasses and maps are shuffled to any check in the game. As with the Map/Compass mode, the item tracker is launched with “?” images for the dungeon prizes. The user must change the dungeon prize.

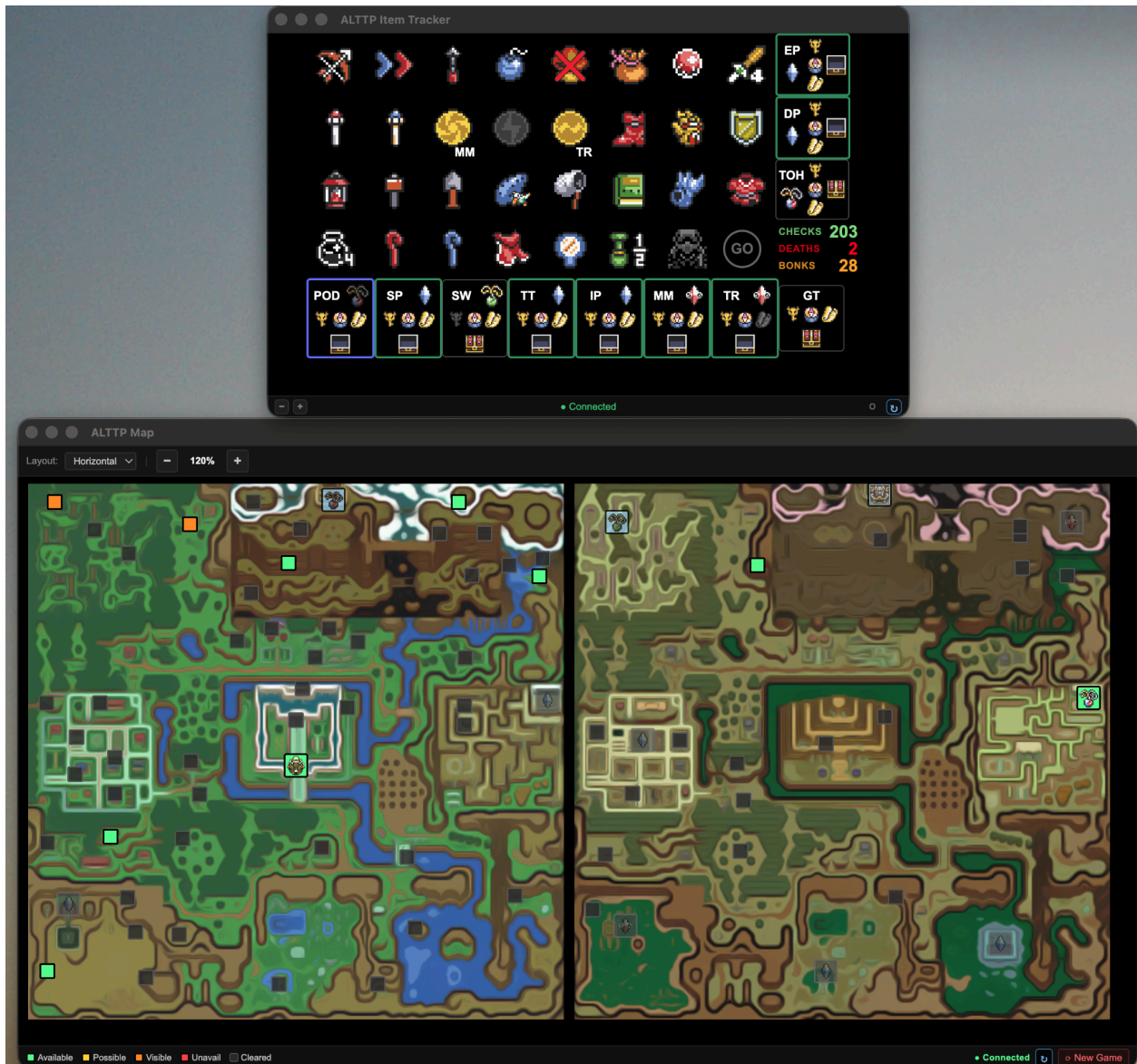
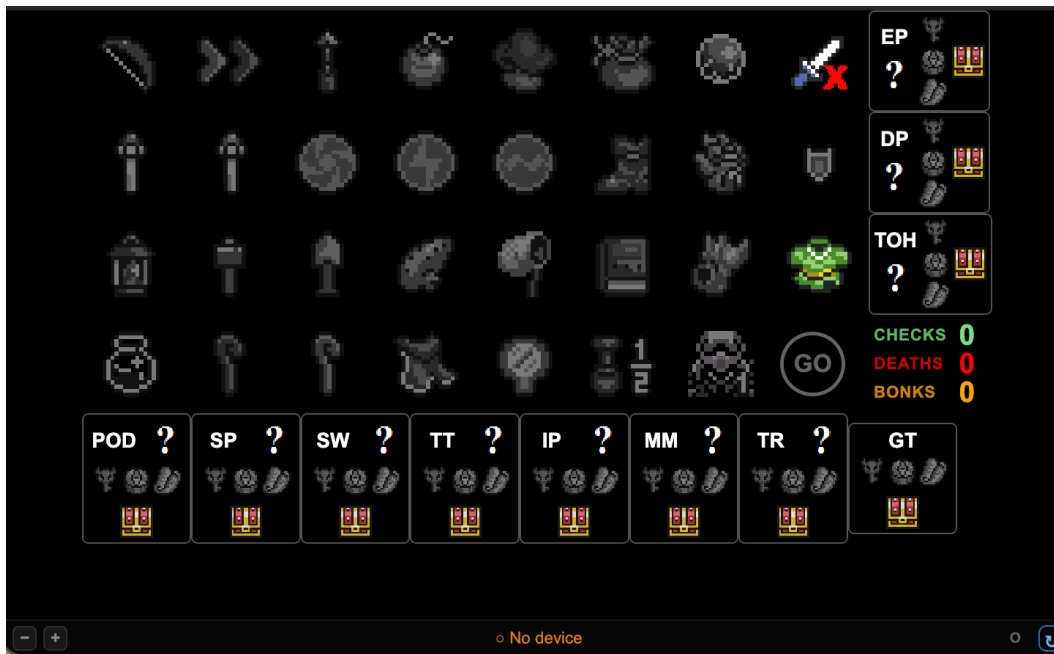




Other Mode

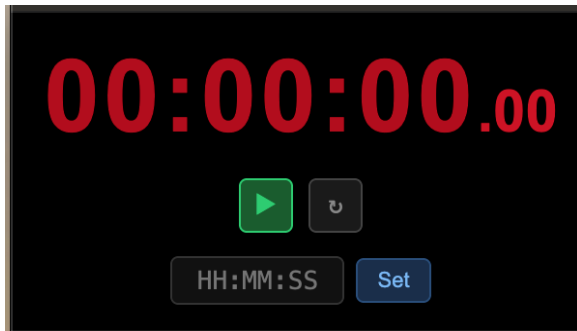
“Other” mode is to accommodate the other types of seeds, such as dungeon room shuffle. The item tracker will track Crystal/Pendants, as with each other mode, as well as Big key, Map and Compass for each dungeon. The dungeon items are not counted. The user will need to click on the chest icon to mark the dungeon items as complete. As with the Map/Compass mode, the item tracker is launched with “?” images for the dungeon prizes. The user must change the dungeon prize.

NOTE: There is no plan on implementing entrance shuffle or shop sanity. Other trackers exist and implementation would take many hours. Sorry, not a mode that I want to tackle.

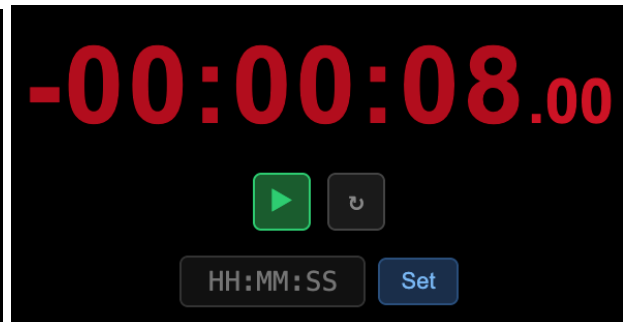
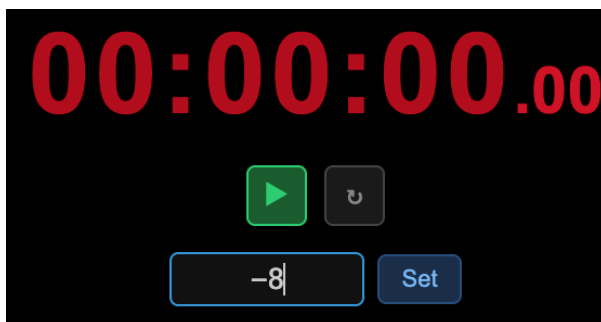


Using the Timer

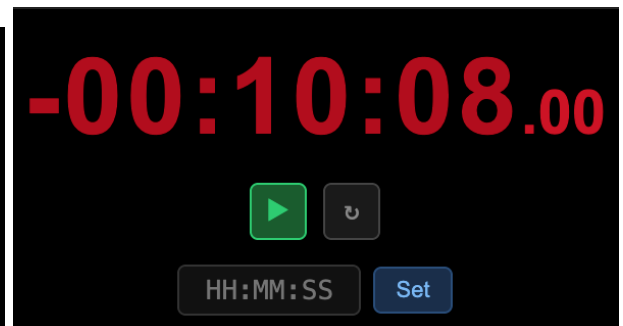
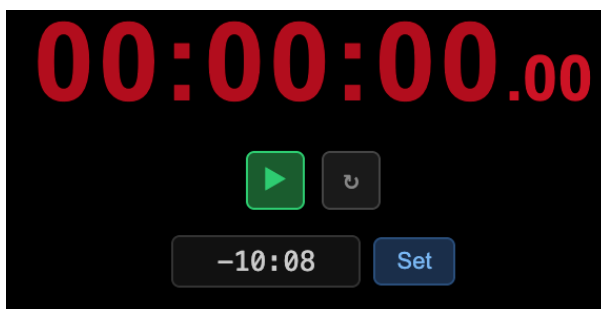
The timer is an option for players to track time. Functionality is fairly simple.



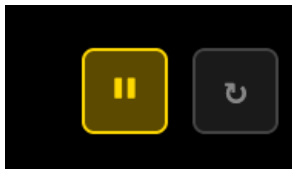
The timer supports negative time to count up if desired.
Set the time by -8

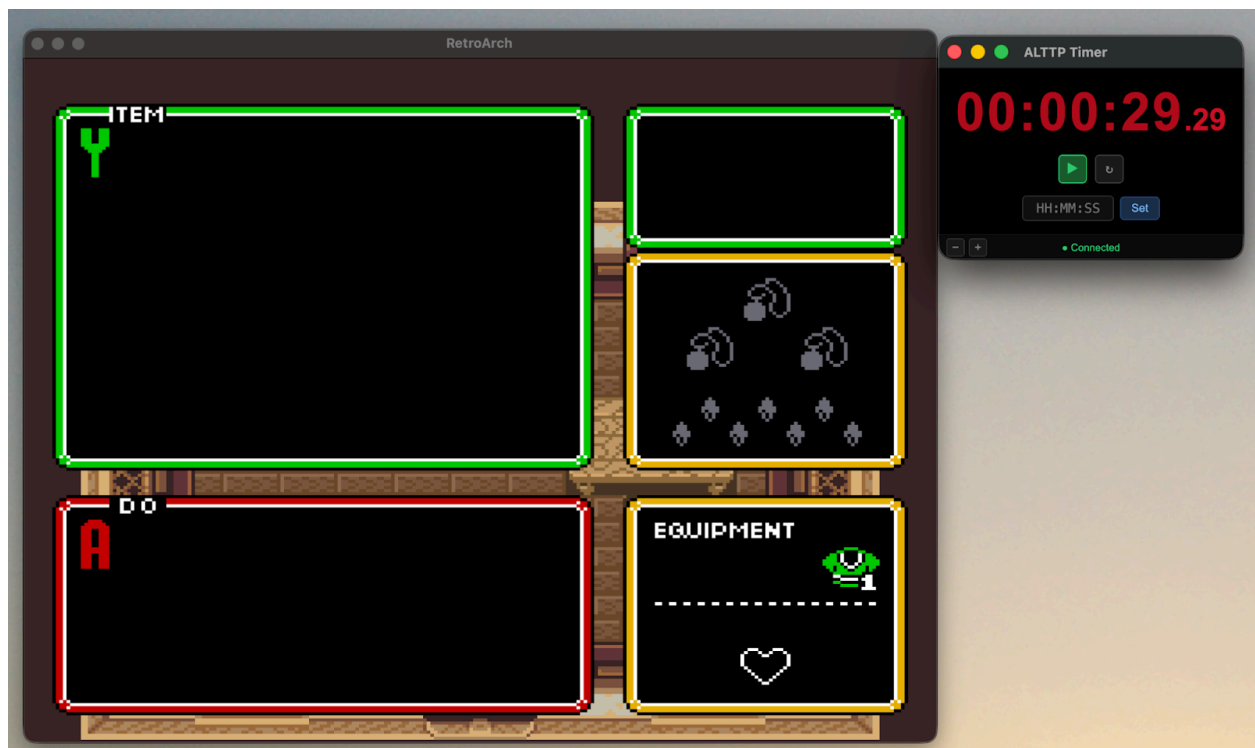


Minutes will work as well

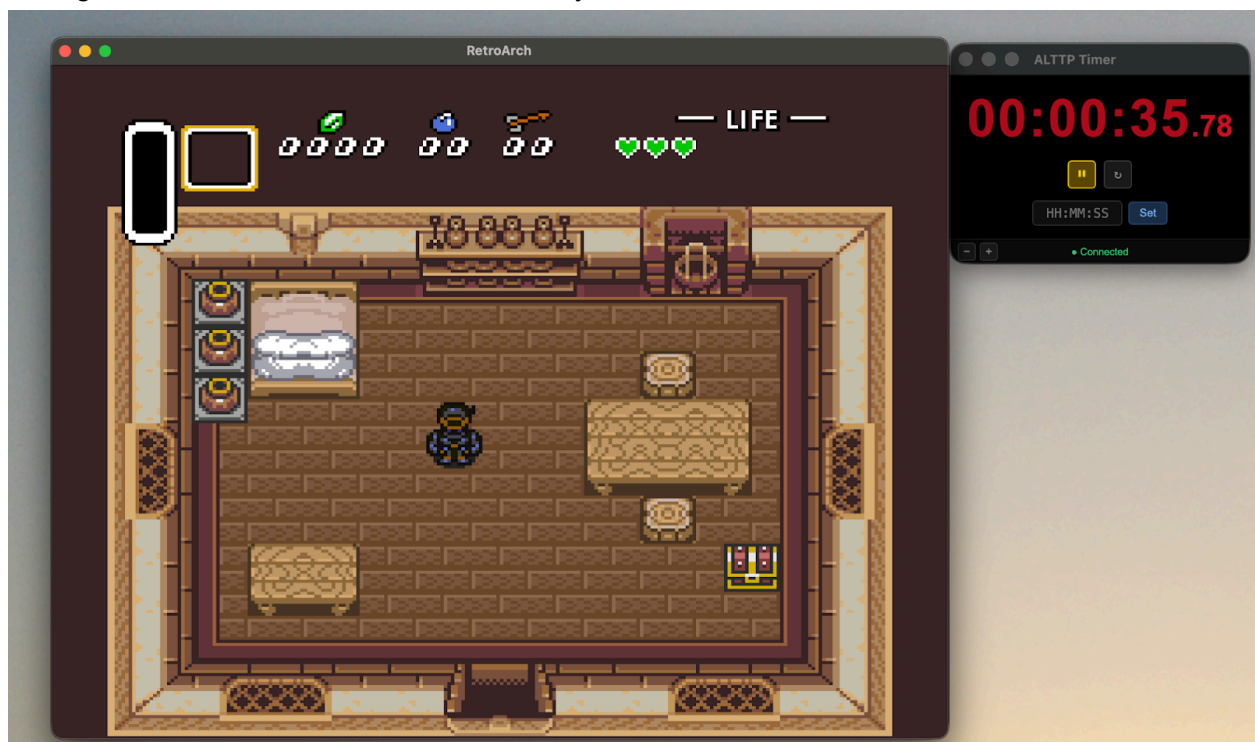


Once connected, the timer will start when the game starts. The timer can be paused when the game is in a menu. This is done by pressing the pause button.





Exiting that menu, the timer will automatically start.



NOTE:

It is a bit difficult to capture every possible scenario when to pause the game, but pausing will work from any menu. It will start during game play. The timer is programmed to stop once the Triforce room is entered. This is completing the game.

Required Software Installation

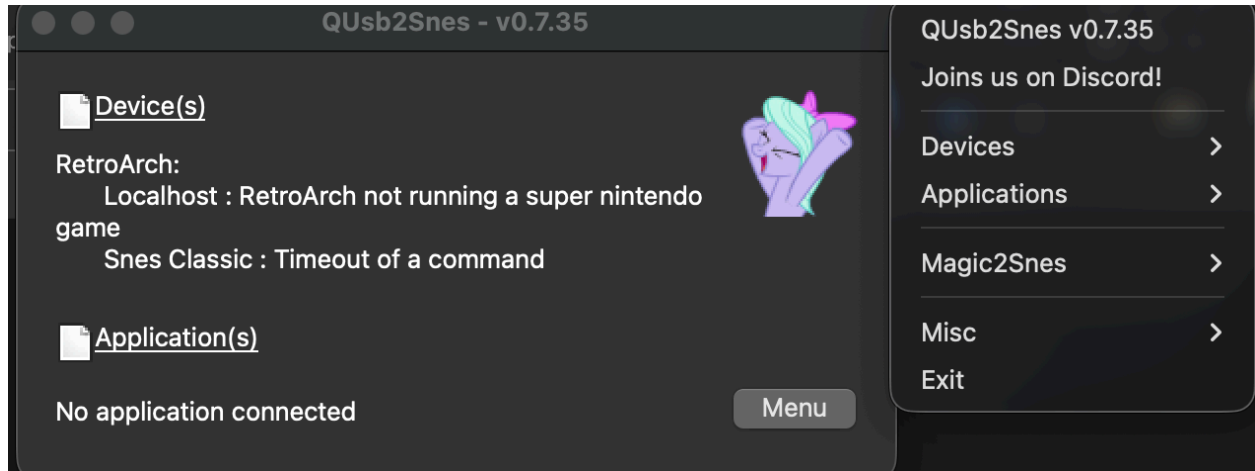
Retroarch 1.2.2 or greater is preferred.

Install either QUSB2SNES or SNI. Only one program can operate on that port at a given time.

QUSB2SNES Setup

Install QUSB2SNES as per <https://skarsnik.github.io/QUsb2snes/>

NOTE: Windows may require Microsoft Visual C++ redistributable from Microsoft.



Windows will look like this when launched Select RetroArch:

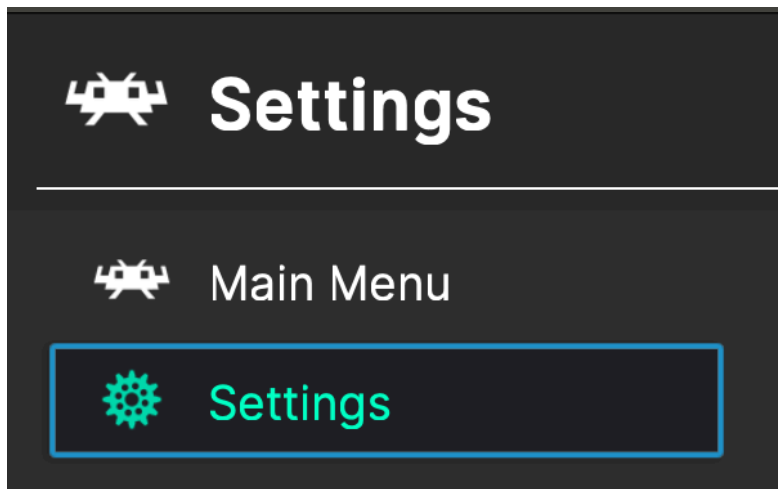
Select a Device

Please select the type of device you will use with QUsb2Snes

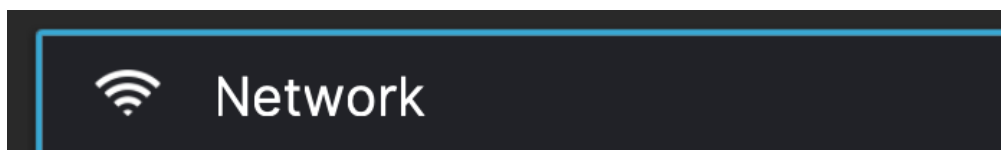
- ☐  A Physical console (SNES, SFC, SuperNT) with a Sd2Snes or FxPak pro.
- ☒  The RetroArch Emulator
- ☐  Emulator with Emulator Network Access support
- ☐  SNES Classic mini

RetroArch Setup (QUSB2SNES)

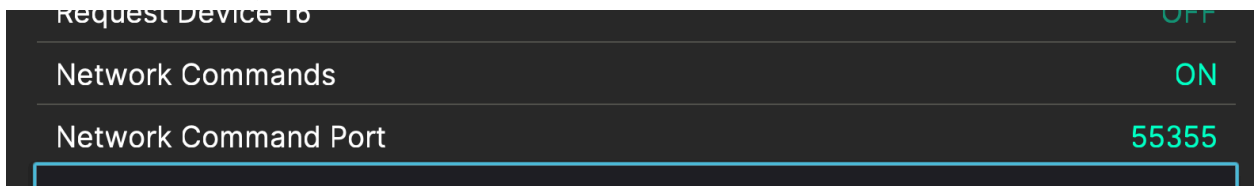
1. First setup the network.



2. Select Network

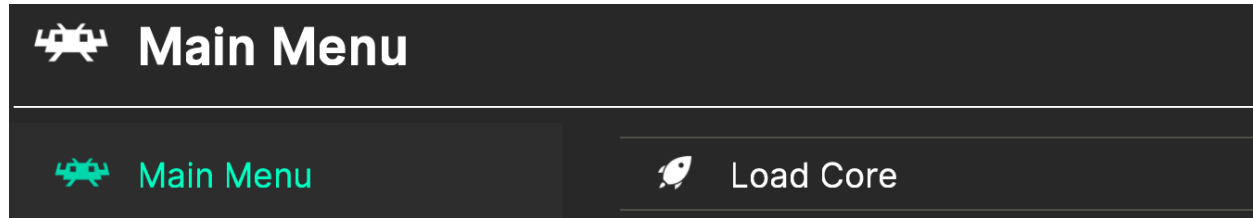


3. Scroll to the bottom, and set the Network Commands to ON

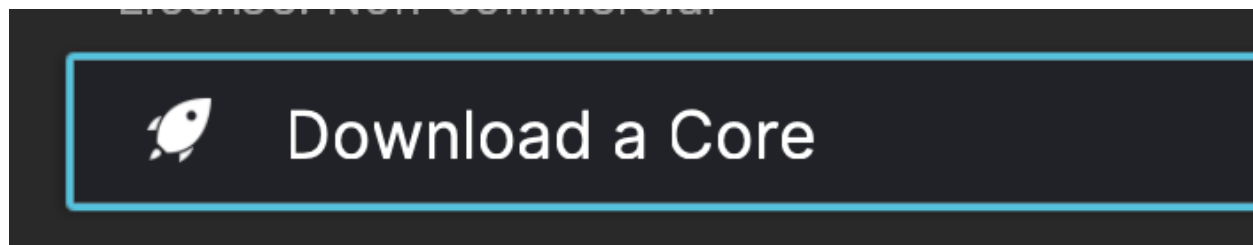


4. Install the BSNES-Mercury Core

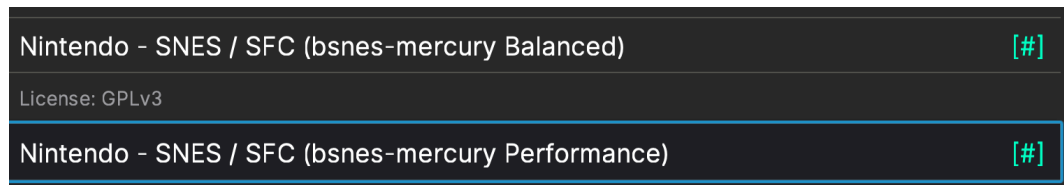
1. Navigate to the Main Menu, and Select Load Core



2. Select Download a Core



3. Search for SNES and look for the bnes-mercury Core



- 4.

NOTE: May need to test both Performance and Balanced to see what works better for you

SNI Install

WINDOWS

Download the latest from <https://github.com/alttpo/sni/releases>

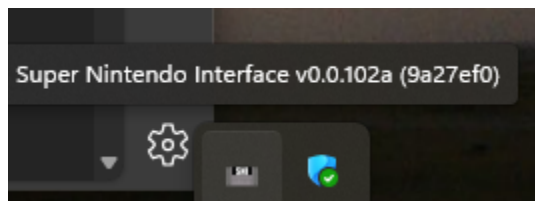
NOTE: Most common is the *Windows-amd64.zip*

Extract the .zip

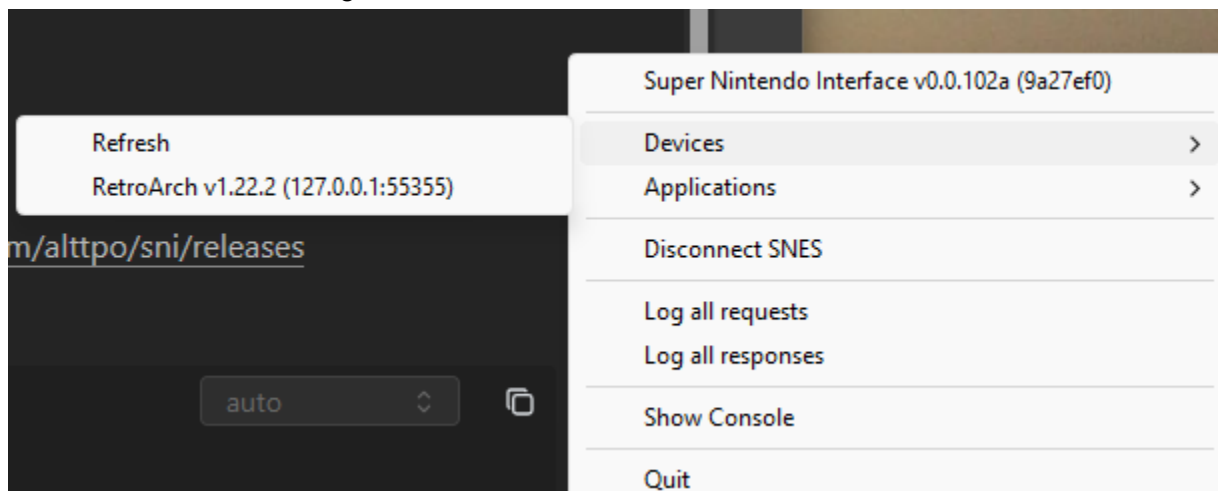
Launch the sni application

sni	3/4/2026 7:36 PM	Application	18,148 KB
sni.proto	3/4/2026 7:36 PM	PROTO File	13 KB

In the taskbar there should be an icon



When RetroArch is running it will show under the devices

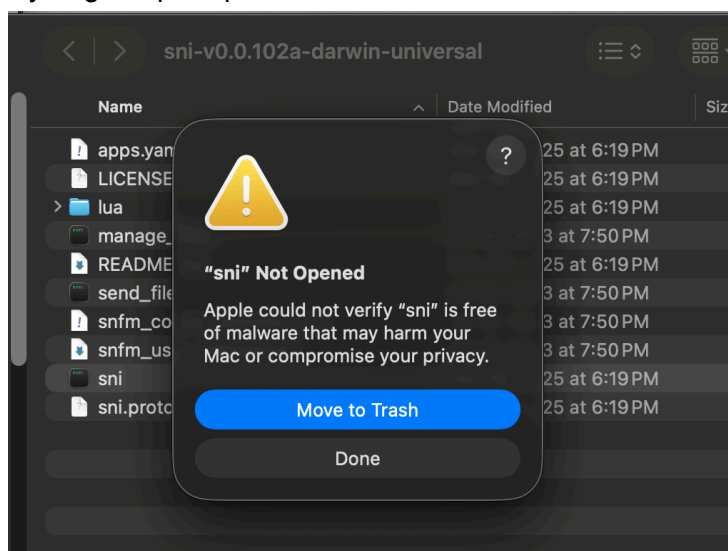


MACOS

Download the “darwin universal” from <https://github.com/alttpo/sni/releases>

Extract it and click the sni

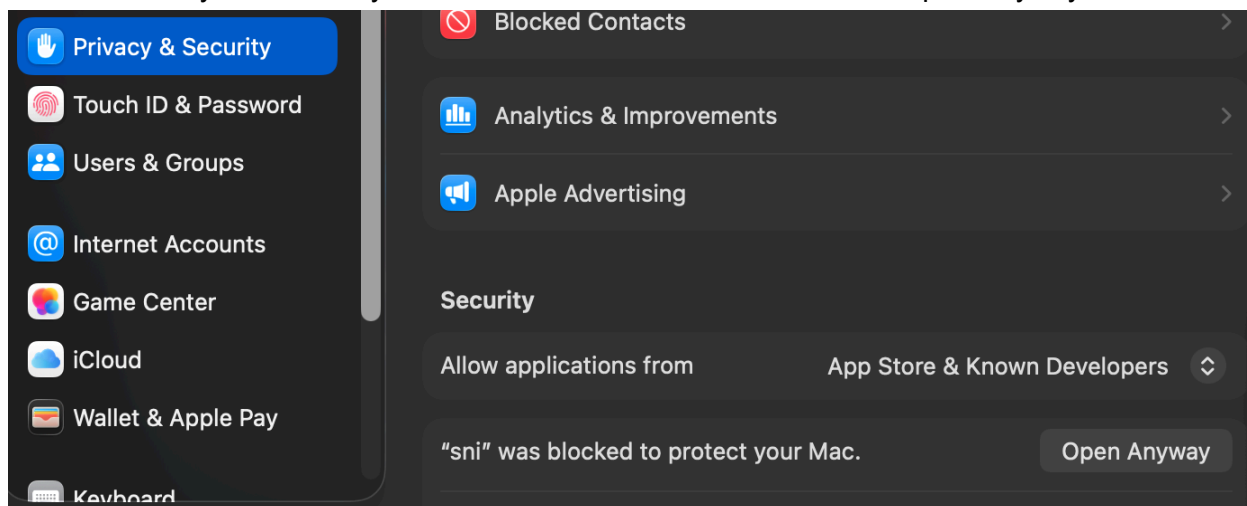
If you get a prompt as shown, select Done



Open Settings from application



Scroll to Privacy and Security. Look for “sni” was blocked and select “Open Anyways”

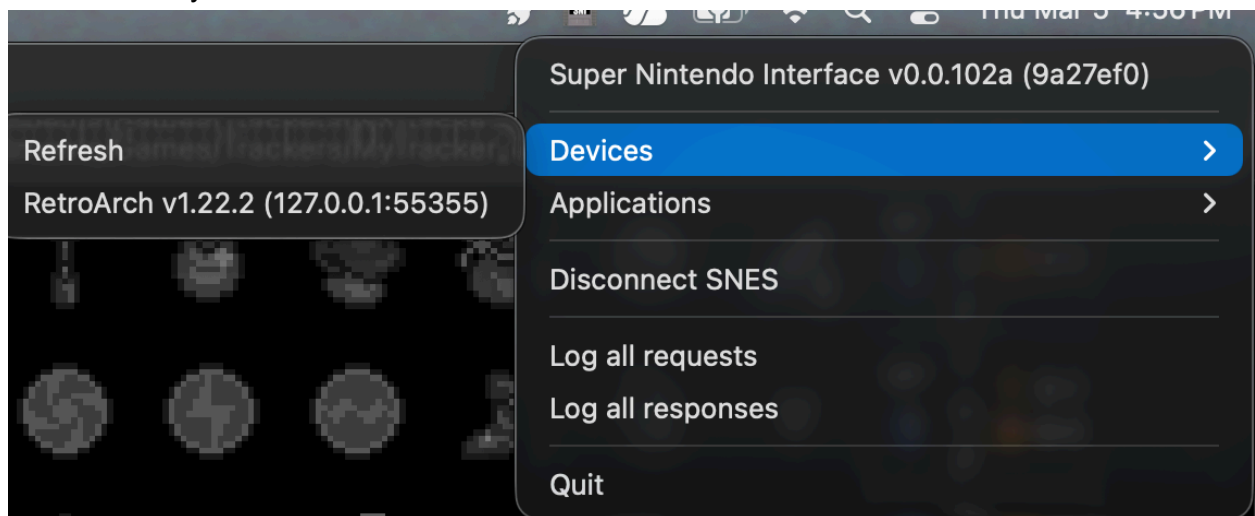


it will launch a terminal window

NOTE: It may work better if launched after opening RetroArch and loading the game.

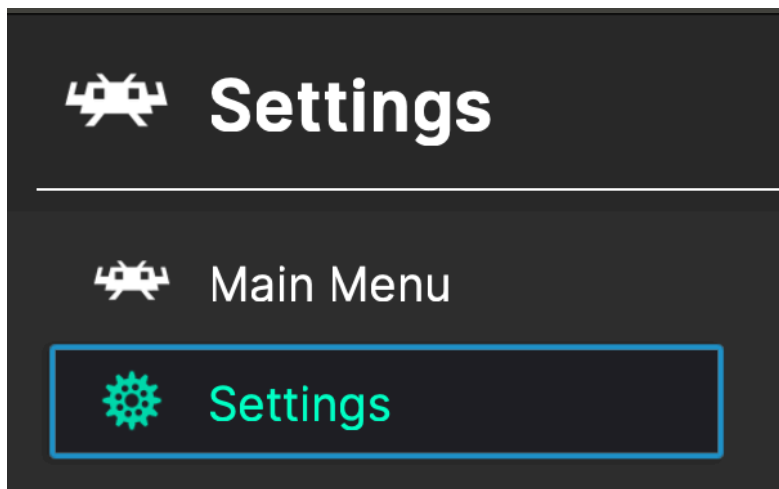
```
00000030 01 00 04 00 b7 ff b7 ff 2c 82 ab 98 00 80 af 98 |.....,.....|
00000040 ff ff ff ff b7 ff 2c 82 2c 82 2c 82 00 80 d8 82 |.....,.....|
2026/03/05 03:05:21.325433 detect: read {address:(FxpakPro $40ffb0 ExHiROM),size
:$50}
2026/03/05 03:05:21.341892 detect: read {address:(FxpakPro $40ffb0 ExHiROM),devi
ceAddress:(SnesABus $40ffb0 ExHiROM),size:$50} complete: score=21
00000000 01 8d 24 01 e2 30 6b 5c 9c b1 a1 ff ff ff ff ff |..$.0k\.....|
00000010 56 54 20 44 76 78 6e 44 36 62 4e 47 61 20 20 20 |VT DvxnD6bNGa|
00000020 20 20 20 20 20 30 02 0c 05 00 01 00 8b 9f 74 60 |      0.....t`|
00000030 01 00 04 00 b7 ff b7 ff 2c 82 ab 98 00 80 af 98 |.....,.....|
00000040 ff ff ff ff b7 ff 2c 82 2c 82 2c 82 00 80 d8 82 |.....,.....|
2026/03/05 03:05:21.342023 detect: read {address:(FxpakPro $40ffb0 SA1),size:$50
}
2026/03/05 03:05:21.358316 detect: read {address:(FxpakPro $40ffb0 SA1),deviceAd
dress:(SnesABus $01ffb0 SA1),size:$50} complete: score=4
00000000 a4 8f aa c1 7e 6b ad e0 02 d0 06 af 56 f3 7e d0 |....~k.....V.~.|
00000010 17 af 57 f3 7e f0 03 9c e0 02 a9 0c 85 4b a9 2a |..W.~.....K.*|
00000020 a6 1b f0 02 a9 14 85 11 6b a9 0c 85 4b a9 2a a6 |.....k....K.*|
00000030 1b f0 02 a9 14 85 11 22 eb 89 a4 38 e9 08 22 b9 |....."....8..."|
00000040 b2 a0 c9 a8 90 06 a9 00 8f 6d f3 7e 6b ff ff ff |.....m.~k...|
2026/03/05 03:05:21.358419 detect: map mode = 30; masking off slow vs fastrom =
20
2026/03/05 03:05:21.358446 detect: detected mapping mode = LoROM
```

In the taskbar you will see the SNI icon. This allows some control over SNI

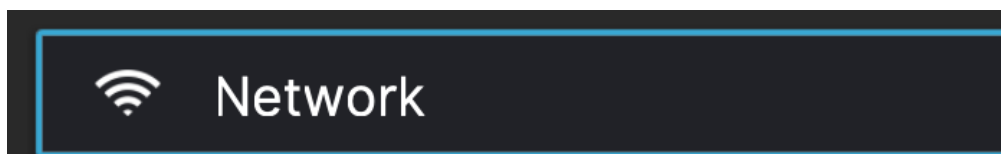


RetroArch Setup (SNI)

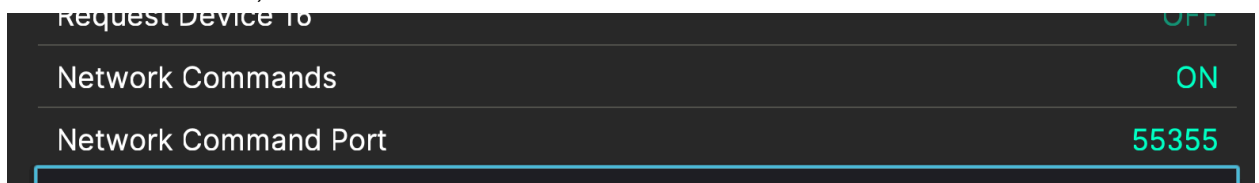
1. First set up the network.



2. Select Network

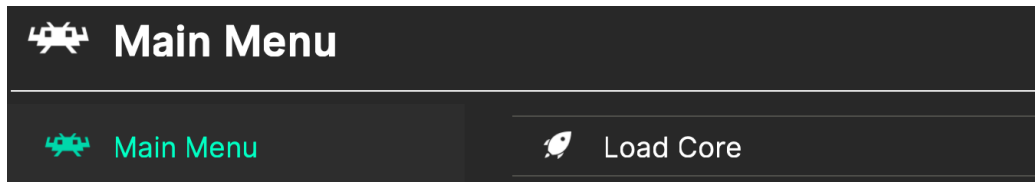


3. Scroll to the bottom, and set the Network Commands to ON

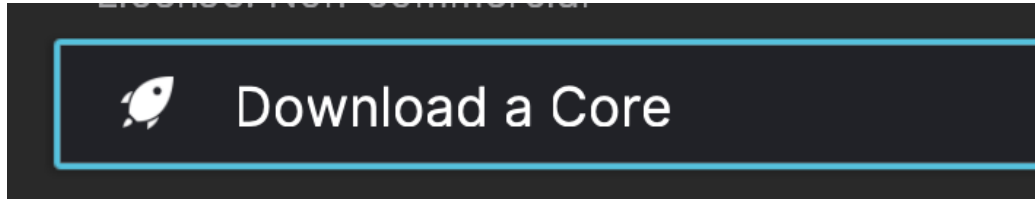


4. Install the "bsnes Mercury" Core

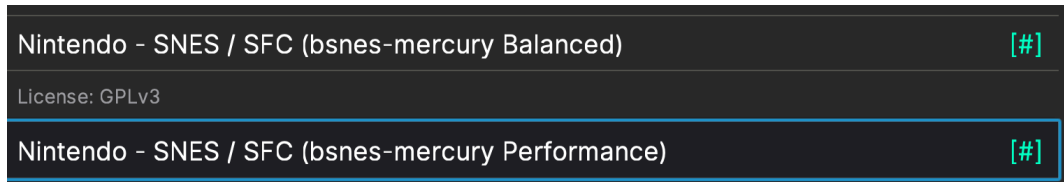
- a. Navigate to the Main Menu, and Select Load Core



- b. Select Download a Core



- c. Search for SNES and look for the “bsnes-mercury Balanced” or “bsnes-mercury Performance” Core



NOTE: May need to test both Performance and Balanced to see what works better for you.

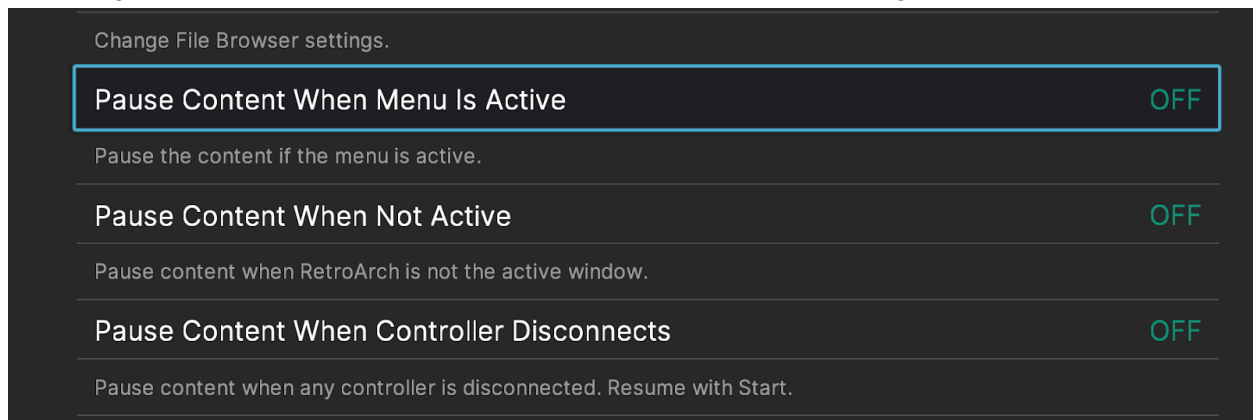
Troubleshooting

With almost all trackers, it is best to launch the software in the proper order

- Launch QUSB2SNES or SNI
- Open the Emulator (RetroArch)
- Launch the game to the title screen
- Launch the various tracker components (map, Items, timer)
 - These should all show connected
- Start the game

Retroarch pausing may affect the timer functions.

In Settings under User Interface and turn off the pause content settings as shown below.

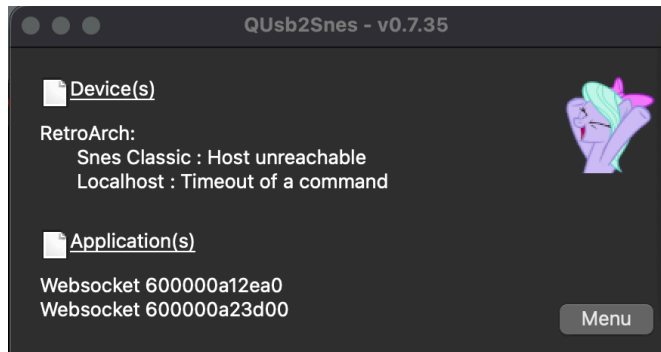


Tracker is not connecting

Example 1:

To troubleshoot, open the websocket interface from the taskbar. This will provide further information..

In this example the Websocket is showing 2 Applications connected (ItemTracker and Map), but RetroArch is not running.



Solution Launch RetroArch

Example 2:

QUSB2SNES or SNI are not running.

Solution Run the software.

Example 3:

“I have tried the above solutions without success”

1. Close the applications.
 - a. Ensure SNI/QUSB2SNES are not running
 - b. Quit RetroArch
 - c. Close the Tracker
2. Launch in order
 - a. Launch QUSB2SNES or SNI
 - b. Open the Emulator (RetroArch)
 - c. Launch the game to the title screen (Ensure using the proper Cores for the websocket
 - i. QUSB2SNES = BSNES-Mercury Balanced or Performance (SNES9x works but deprecated)
 - ii. SNI = BSNES-Mercury Balanced or Performance
 - d. Launch the various tracker components (map, Items, timer)

Example 4:

“Nothing works”

Uninstall the QUSB2SNES or SNI. Reinstall.